AR-Mini · System Architecture v3

As built on 28.10.2026: an Olimex ESP32-S3-DevKit-Lipo runs the robot, the XIAO is the camera node · hardware KILL on the track-driver enable · power fused per branch · dashed = designed or on order

Power



Turnigy 3S 2200 mAh
11.1 V · 40 C · XT60

12 V



Inline fuse 30 A
GT4652 · on the XT60 lead

12 V



Main switch 50 A
R13-401A2A · cuts everything

12 V



6-slot fuse block
branches below · card C026

Fused branches (12 V)

- F1 20 A — thrusters: ESC ×2 (T200 next)
- F2 15 A — servo buck → 6.0 V for the cradle
- F3 10 A — BTS7960 ×2 → tracks
- F4 5 A — logic buck → 5 V + VBAT sense



MZHOU buck
12 → 5 V 3 A · USB-C

F4 5 A into the DevKit OTG

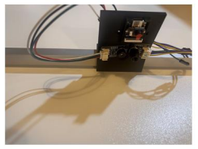


Bench today
Jesverty 12 V · CC limit

Every test so far ran from the lab supply; the LiPo chain is the next card.

Charged off-robot on SkyRC iMAX B6 V2.
Pi 5 rail (QIQIAZI buck) only in v2.

Sensing



Camera head on the arch
printed mount · 3 cameras · C047



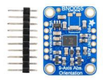
MLX90642-D45
45° · fwd · C048



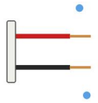
MLX90640 110°
wide · C029



OV3660 2 MP
XIAO · Wi-Fi



BNO055 IMU
not wired yet



Leak probe
GPIO7 · wired

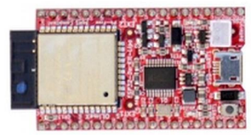


VBAT divider
100k / 30k · C027

I²C-1 · I²C-0 · I²C-0 → PCA9685 · VBAT · LEAK

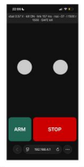
Control

GPIO4 VBAT · GPIO7 LEAK · GPIO14 KILL (EN node) · GPIO8 / 9 I²C-1 · GPIO48 / 47 I²C-0 · USB-C OTG 5 V in · USB-C UART laptop



Olimex ESP32-S3-DevKit-Lipo
two BB400 breadboards · PlatformIO

GPIO1 / 2 ESC_L / ESC_R · GPIO39–42 BTS7960 PWM · Wi-Fi AP "AR-Mini" phone



Phone remote
192.168.4.1 · 500 ms heartbeat



XIAO ESP32-S3 Sense
camera node · own AP "AR-Mini-cam"

Firmware · PlatformIO
5 envs · bench, remote, cam



S8050 stage
open collector · 5 V

Mission state machine (firmware)

SEARCH → APPROACH → ALIGN → LIFT → RETURN

any state → SAFE on KILL · LEAK · LOW BATTERY · lost heartbeat · 2 s dead-man (thrusters 1500 µs, tracks 0, cradle holds)

Three stops in series on the motors: the rocker cuts the BTS7960 enable in hardware, 10 k pull-downs hold GPIO39–42 low through boot (C015), and the firmware drives every output to neutral. Tested: KILL stops the thruster within 40 ms (C025) and the track at once (C050).

Logs: CSV over USB on the bench (C027 calibration, C050 currents) · 54 loop cards in docs/LOG.md · FAILS_AND_FIXES.md 177 entries

Next · own PCBs, level shifter, v2 (2027)



XIAO breakout rev 0
built · AISLER



Carrier rev B
designed · C032



×5

SN74AHCT125N
3.3 → 5 V · to buy



×3

WROOM-1 N16R8
on the carrier



Raspberry Pi 5
v2 2027 · ROS 2



Active cooler
v2 2027

Actuation & rescue



KILL rocker · SPDT
C = EN · RUN 3V3 · KILL GND

EN · F3 10 A



BTS7960 L
track L · C050 PASS

EN · F3 10 A



BTS7960 R
track R · C051 PASS

M+ M−



ROB0154 tracks
built 25.10 · 2 × DC 12 V

also → GPIO14

Tracks: 0.08 A at 20 %, 0.34 A at 80 % on the bench (12 V, free-running) · VCC 3V3 so the 3.3 V PWM is clean (FAILS 165)

F1 20 A



Cuifati 560 KV · left
works · 0.58 A · C025/C028



Cuifati · right
faulty · FAILS 173

F1 20 A



×2

Blue Robotics T200
ordered 26.10



×2

Basic ESC R3
1500 µs stop · 3.3 V signal

GPIO1 → S8050 → ESC signal (fw inverts the duty)



Kick-up transom mount
Ø5 hinge · C041/C042



MG90S rudder (interim)
GPIO2 · next card



First float · 24.10
draft 20 mm · dry hulls · C028



Pesho · thermal dummy
red PLA · 37 °C · C021

F2 15 A → 6 V



CH10/11

PCA9685
0x40 · CH10/11 mirrored · 6 V



×2

TD-8120MG
cradle lift · 50 Hz

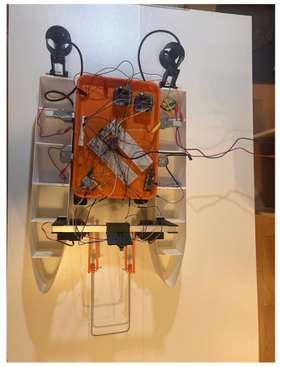


Parallelogram arms
Ø5 cradle U · C023



×4

ST3215 bus servo
option B · spare



The robot as built · 28.10 (C053)
hulls, beams, tray, arch, cradle, tracks, thrusters

What is tested (card → PASS):

C023 cradle lifts Pesho (mirrored servo pair)
C025 thruster L at 12 V, KILL → 1500 µs in 40 ms
C027 VBAT reading ±0.02 V from 10 to 12.6 V
C028 whole robot floats, phone drive over Wi-Fi
C047–C049 camera head: D45 sees a 39.8 °C head at 1.3 m
C050 / C051 both tracks run from the BTS7960s
C053 first dry assembly of every subsystem

Open before G1 (30.10):

right thruster: interim rudder, then 2 × T200
full rescue cycle in water (P22)
power chain on the LiPo (C026)
tray mount (today: a book as a spacer)

Wires: 12 V (fused branch) · 6 V servo rail · 5 V · I²C-0 (Wire) · I²C-1 (Wire1) · PWM / ESC signal · analog · KILL / EN · motor power · USB · Wi-Fi

Dashed card = designed, on order or not in the loop yet. Solid wire = wired and tested on the bench (card numbers = tests in docs/LOG.md). Common GND everywhere; 12 V never enters the breadboards.

AmphiReach · AR-Mini 1:9 demonstrator · architecture v3 (as built 28.10.2026; v1 06.10, v2 27.10 in docs/diagrams/) · photos: BOM listings + build cards · T. Kyoseva, Sofia